用 Nix 打造
可重現的 Linux 桌面

willie

Outline

- Nix 基礎
- Linux Ricing

https://devvillie.me/blog/hackersir-nix/

Nix 是什麼？

Nix: 套件管理器

NixOS: 系統

套件管理器 (Package Manager)

Debian 系: apt

Arch: pacman

Fedora: dnf

NixOS: nix

Nix 與一般套件管理器有何不同？

- Reproducible: 可重現性
- Declarative: 宣告式
- Reliable: 可靠

Reproducible, Declarative

- Nix 語言

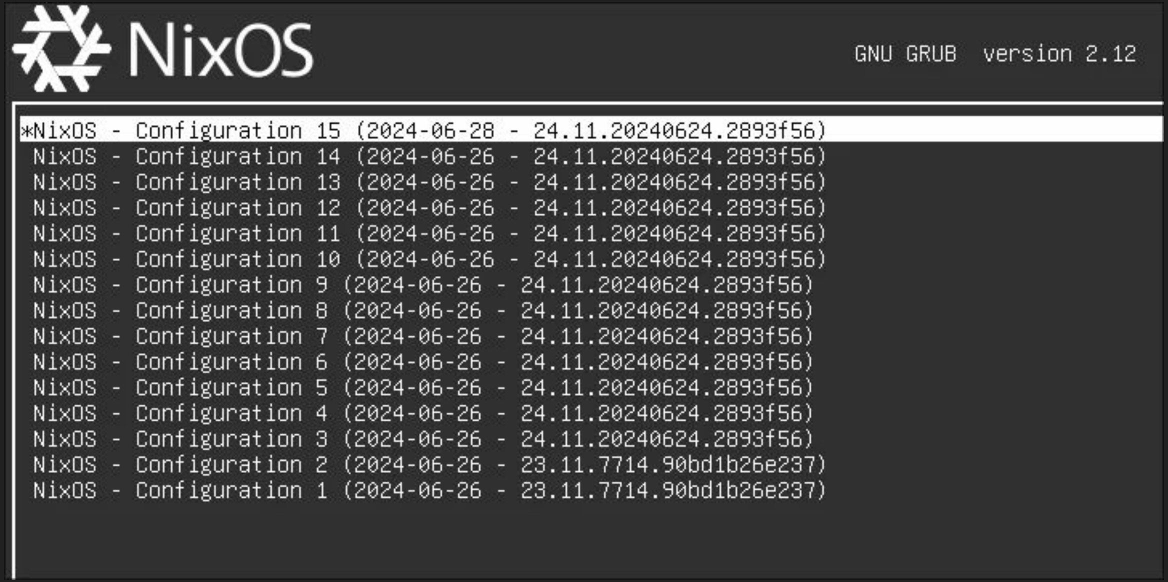
```nix
{
  services.nginx.enable = true;

  users.users.willie = {
    isNormalUser = true;
    extraGroups = [ "wheel" "networkmanager" ];
  };

  environment.systemPackages = with pkgs; [
    git
    vim
    firefox
  ];
}
```

Reliable

grub 可選 generation



Reliable

套件安裝位置

- 其他 Linux
  - /usr/bin
  - /usr/lib
  - /etc
- NixOS
  - /nix/store/abc123-openssl-3.0.13
  - /nix/store/def456-python-3.11.8
  - /nix/store/xyz789-nginx-1.24.0

NixOS 優缺點

優點

- 擁有最多套件
  - https://repology.org/repositories/statistics/newest
- 可快速回滾套件
- 環境可控、乾淨
- 系統可重現

缺點

- 幾乎無法運行非 Nix 的 Binary
- 與一般 Linux 差異很大

Nix 基礎語法

- https://nix.dev/tutorials/nix-language
  - functional programming

Nix 基礎語法

List

```
[
  git
  jq
  python3
]
```

用空白、換行分隔

不用逗號

Nix 基礎語法

變數、String

```
let
  name = "Nix";
in
  "Hello ${name}"
```

let … in … 定義變數

${var_name} 字串內插入變數

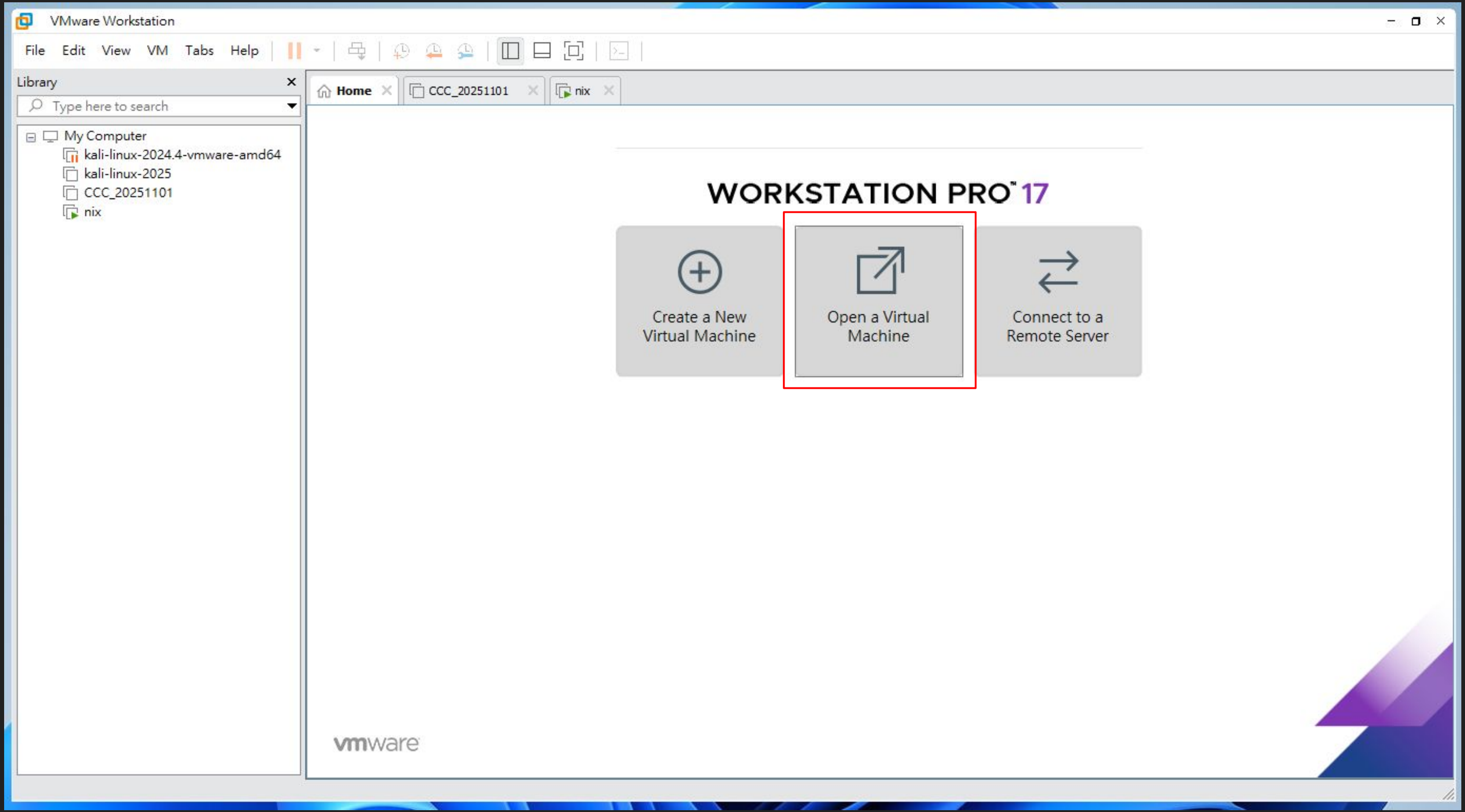
Nix 基礎語法

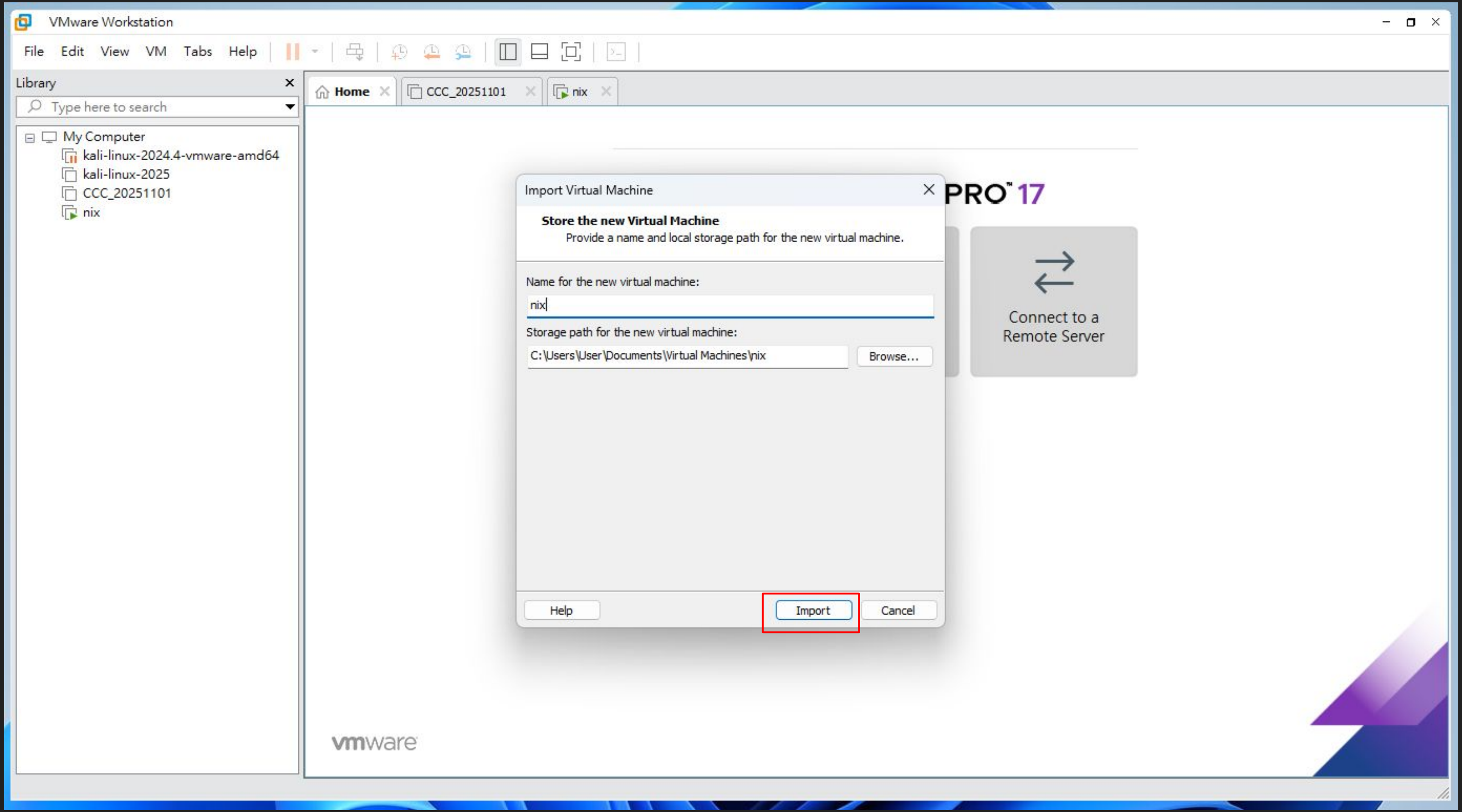
with ...; [ ]

```
packages = with pkgs; [
  git
  jq
  python3
];
```

等同於

pakages = [pkgsg.git pkgs.jq pkgs.python3];


VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2024.4-vmware-amd64
kali-linux-2025
CCC_20251101
nix
Home
CCC_20251101
nix
WORKSTATION PRO 17
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
vmware


VMware Workstation
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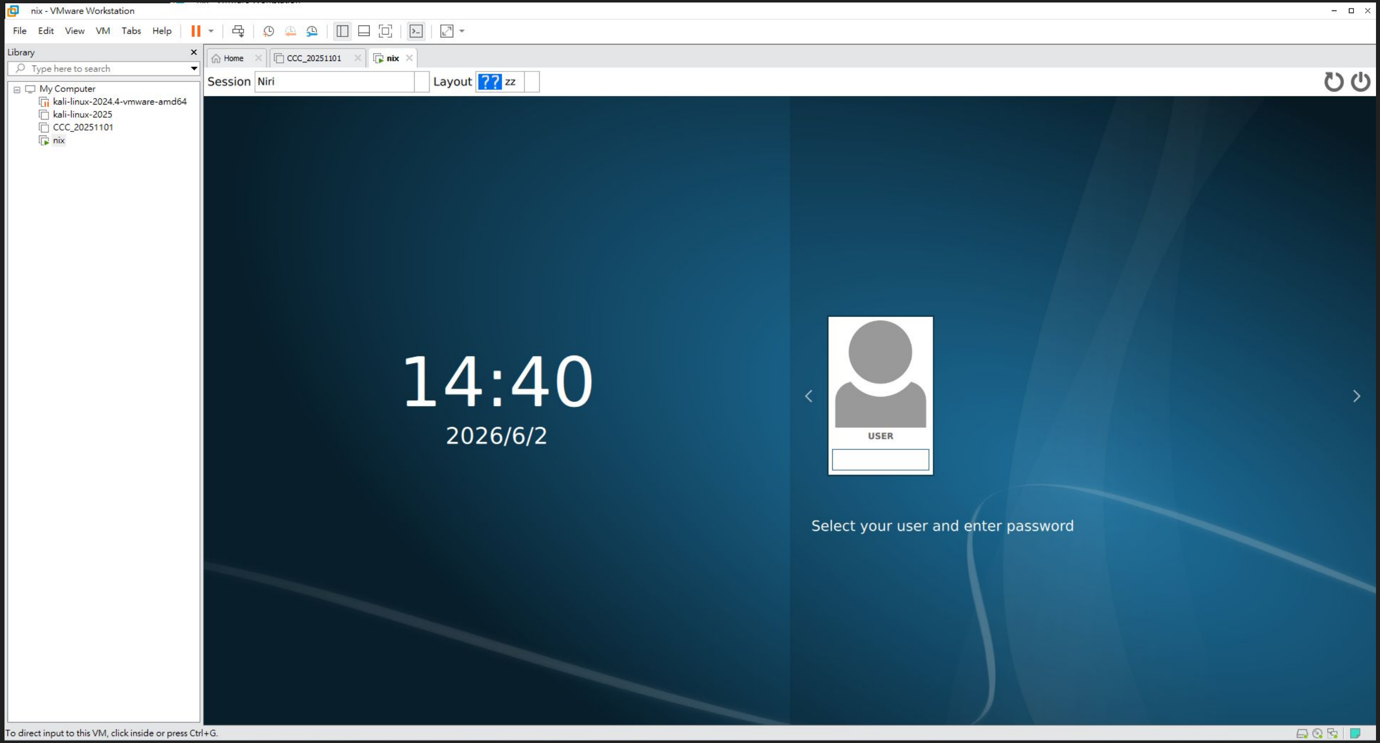
Type here to search
My Computer
kali-linux-2024.4-vmware-amd64
kali-linux-2025
CCC_20251101
nix
Home
CCC_20251101
nix
PRO 17
Connect to a Remote Server
Import Virtual Machine
Store the new Virtual Machine
Provide a name and local storage path for the new virtual machine.
Name for the new virtual machine:
nix
Storage path for the new virtual machine:
C:\Users\User\Documents\Virtual Machines\nix
Browse...
Help
Import
Cancel
vmware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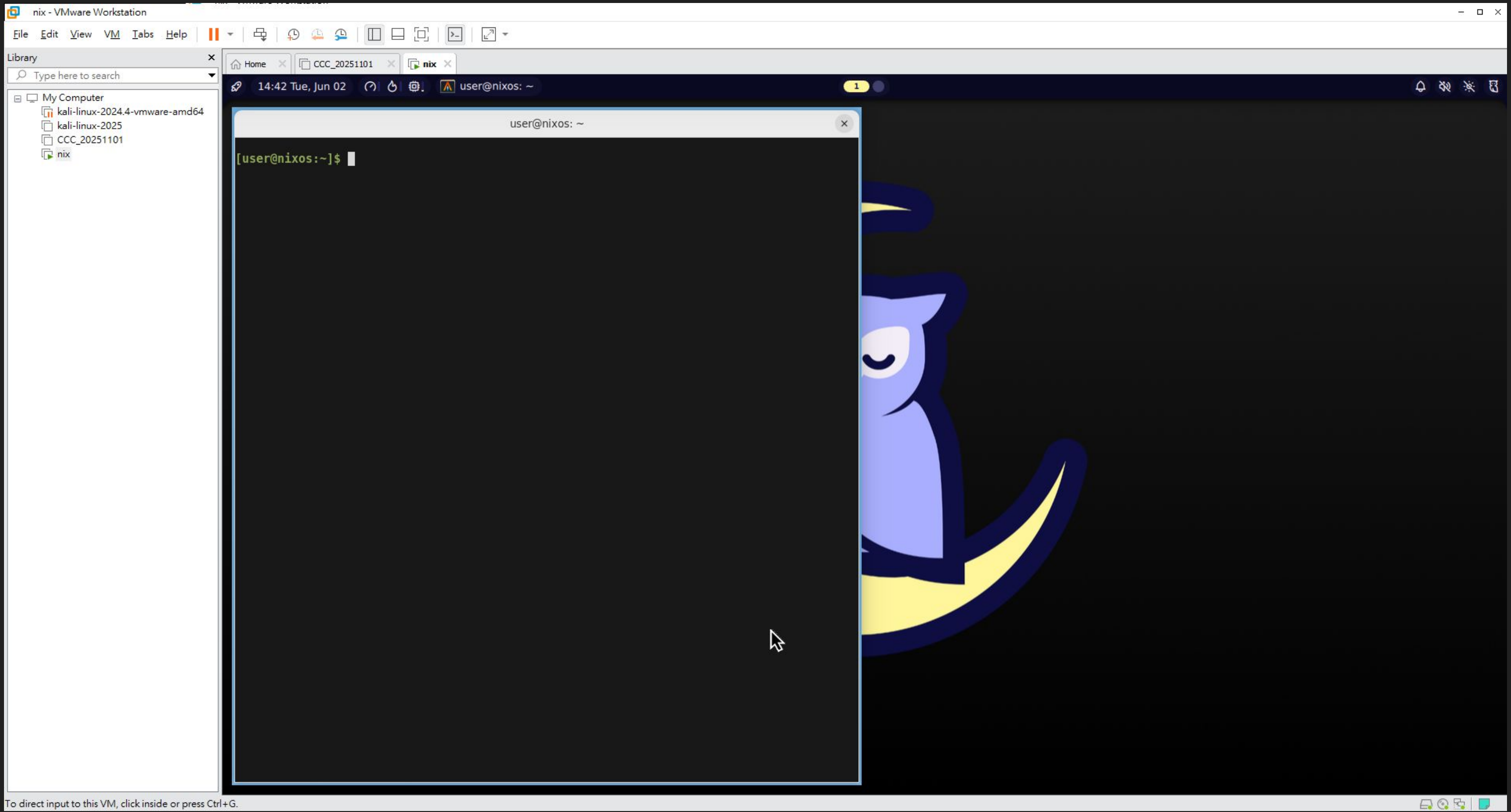
File Edit View VM Tabs Help
Library
Type here to search
My Computer
kali-linux-2024.4-vmware-amd64
kali-linux-2025
CCC_20251101
nix
Home
CCC_20251101
nix
Session Niri
Layout ?? zz
14:40
2026/6/2
USER
Select your user and enter password
To direct input to this VM, click inside or press Ctrl+G.

輸入密碼: 1234



win + t: 開啟 terminal
win + f: 全螢幕
win + 滑鼠左右鍵: 拖曳、縮放視窗

左上角圖標: 開啟應用程式
右上角圖標: 開始選單？idk

sudo vi /etc/nixos/configuration.nix

定義整個系統組成

sudo vi /etc/nixos/configuration.nix

https://search.nixos.org/packages

安裝、解除安裝系統套件

```
environment.systemPackages = with pkgs; [
    vim
    git
    alacritty
    mako
    mesa-demos
    vulkan-tools
    xwayland-satellite
    noctalia-shell
    thunar
    vscode
#      firefox
  ];
```

nixos-rebuild

- sudo nixos-rebuild switch
  - rebuild 目前 configuration 並直接切換至新 generation
- sudo nixos-rebuild boot
  - rebuild 目前 configuration, 在下次開機時才切換至新 generation
- sudo nixos-rebuild list-generations
  - 列出可用的 generation

實作 #1

- 安裝 firefox

https://devvillie.me/blog/hackersir-nix/

開啟 ssh 服務、設定防火牆

```
services.openssh = {
  enable = true;
  ports = [ 22 ];
  settings = {
    PasswordAuthentication = true;
    AllowUsers = null;
    UseDns = true;
    X11Forwarding = false;
    PermitRootLogin = "prohibit-password";
  };
};
```

```
networking.firewall.allowedTCPPorts = [ 22 ];
```

實作 #2

- 開啟 SSH

https://devvillie.me/blog/hackersir-nix/

flake

```
nix.settings.experimental-features = [ "flakes" ];
```

- 放置於 flake.nix
- 定義所有套件的出處
- 於 flake.lock 鎖定 repo 版本

```
{
  description = "A very basic flake";

  inputs = {
    nixpkgs.url = "github:nixos/nixpkgs/nixos-26.05";
  };

  outputs = { self, nixpkgs }: {
    nixosConfigurations.nixos = nixpkgs.lib.nixosSystem {
      system = "x86_64-linux";

      modules = [
        ./configuration.nix
      ];
    };
  };
}
```

實作 #3

- 改為使用 flake

https://devvillie.me/blog/hackersir-nix/

nix shell

- 舊版
  - nix-shell -p cowsay
- 新版
  - nix shell nixpkgs#cowsay

which cowsay

/nix/store/0r80s9lrgi64l53wrid79yymfyhqmqg8-cowsay-3.8.4/bin/cowsay

實作 #4

- 使用 nix shell

https://devvillie.me/blog/hackersir-nix/

direnv

/etc/nixos/configuration.nix

```
programs.direnv.enable = true;
```

sudo nixos-rebuild switch

shell.nix

```
{ pkgs ? import <nixpkgs> {}}:

pkgs.mkShell {
  packages = with pkgs; [
    python3
    hello
  ];
}
```

echo "use nix" >> .envrc
direnv allow

hello
python

實作 #5

- 使用 direnv

https://devvillie.me/blog/hackersir-nix/

home-manager

管理使用者層級設定、套件、dotfiles 與應用程式配置

Linux Ricing

What is Linux Ricing?

- Linux ricing 是指透過客製化桌面環境、視窗管理器、主題、圖示、字型與設定檔，把 Linux 系統調整成更美觀、更符合個人工作流與審美的過程
  - （By ChatGPT）
- https://www.reddit.com/r/unixporn/
- https://www.reddit.com/r/niri/s/AYtG5sS7vs
- https://www.reddit.com/r/unixporn/comments/1txh5ju/

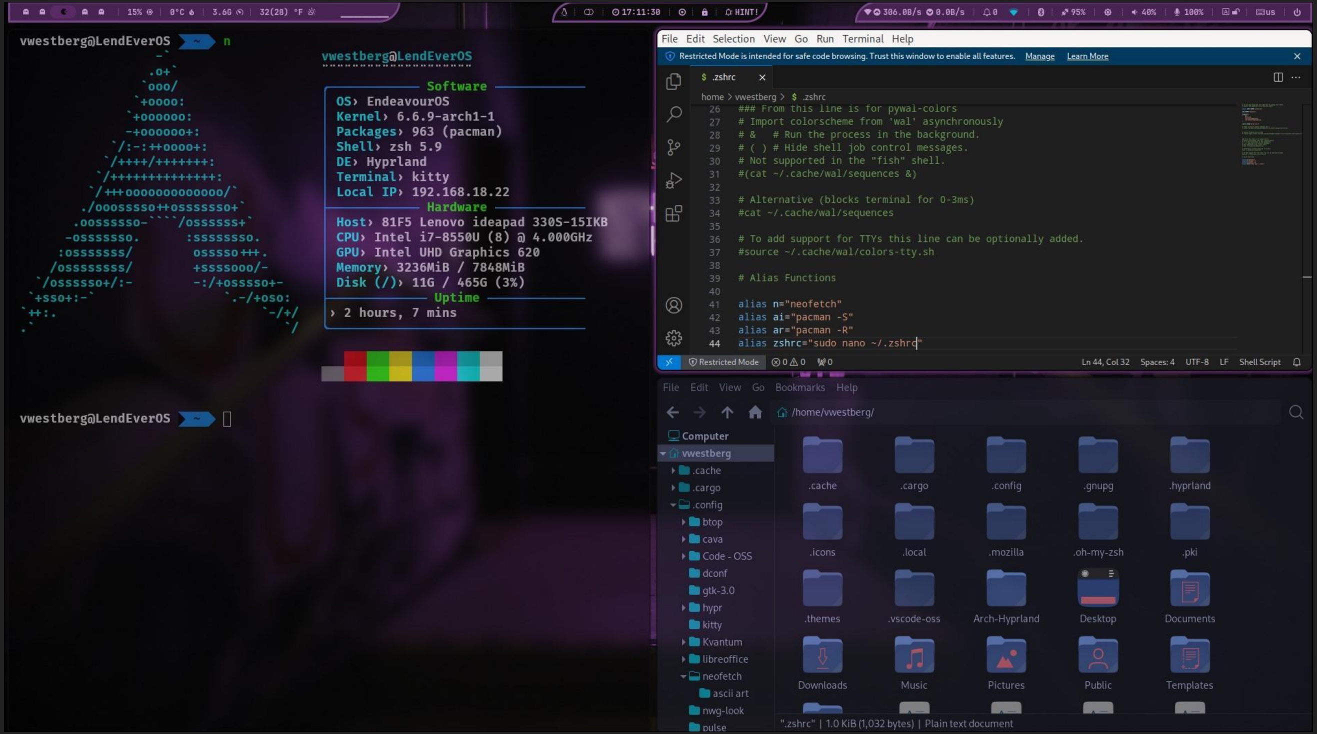
名詞解釋

- DE (Desktop Environment): 桌面環境
  - KDE Plasma: https://kde.org/zh-tw/plasma-desktop/
  - GNOME: https://www.gnome.org/
- WM (Window Manager): 視窗管理器
  - niri: https://github.com/niri-wm/niri (本次使用)
  - Hyprland: https://hypr.land/

名詞解釋

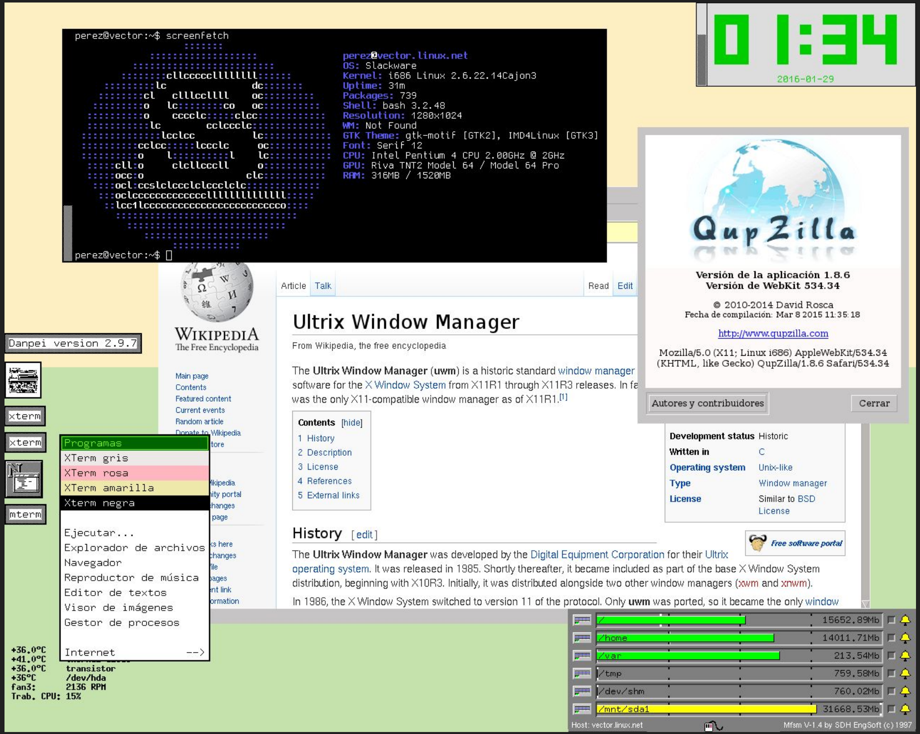
- Tiling Window: 平鋪視窗
  - niri: https://github.com/niri-wm/niri (本次使用)
  - Hyprland: https://hypr.land/
- Floating/Stacking Window: 堆疊視窗
  - Windows、macOS
  - KDE Plasma: KWin: https://userbase.kde.org/KWin/
  - GNOME: Mutter: https://mutter.gnome.org/

名詞解釋



Tiling Window

名詞解釋

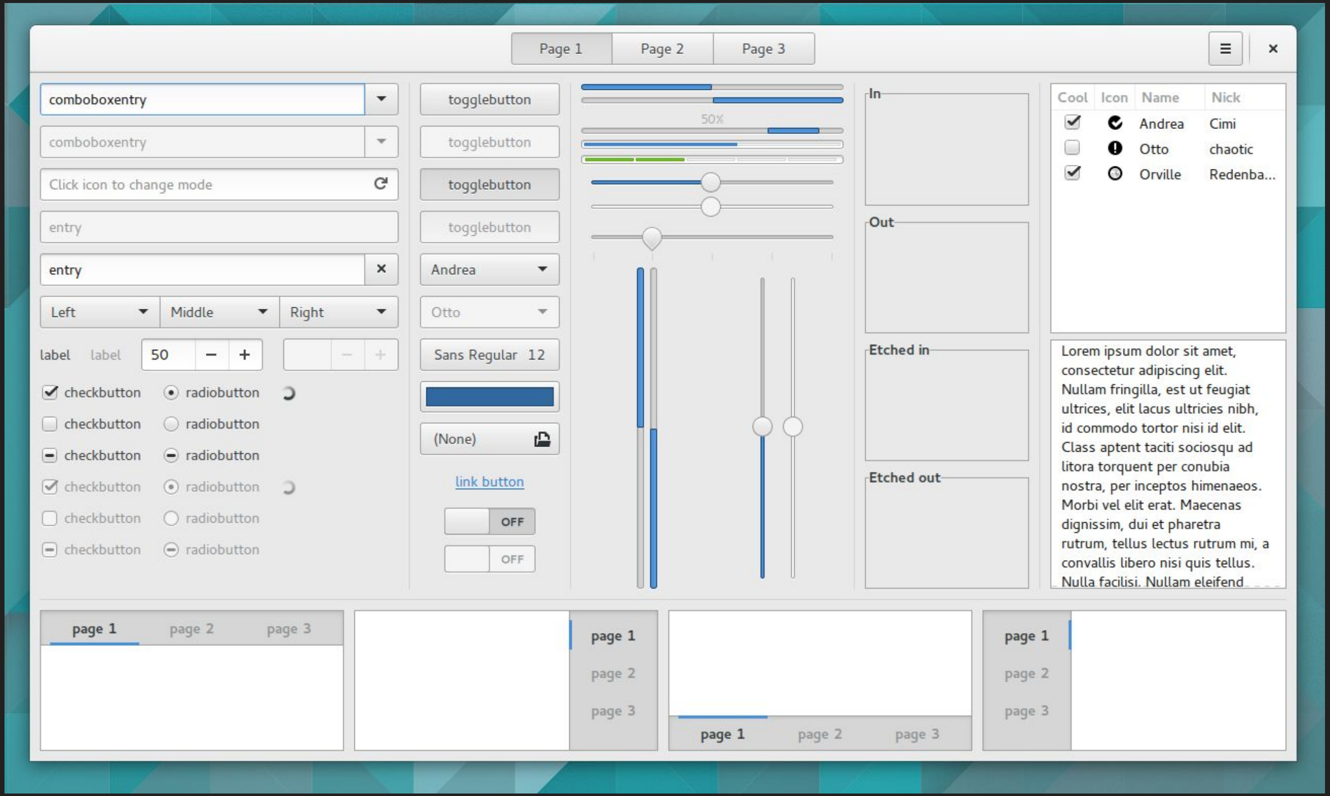


Floating/Stacking Window

GUI 框架

GTK: GNOME

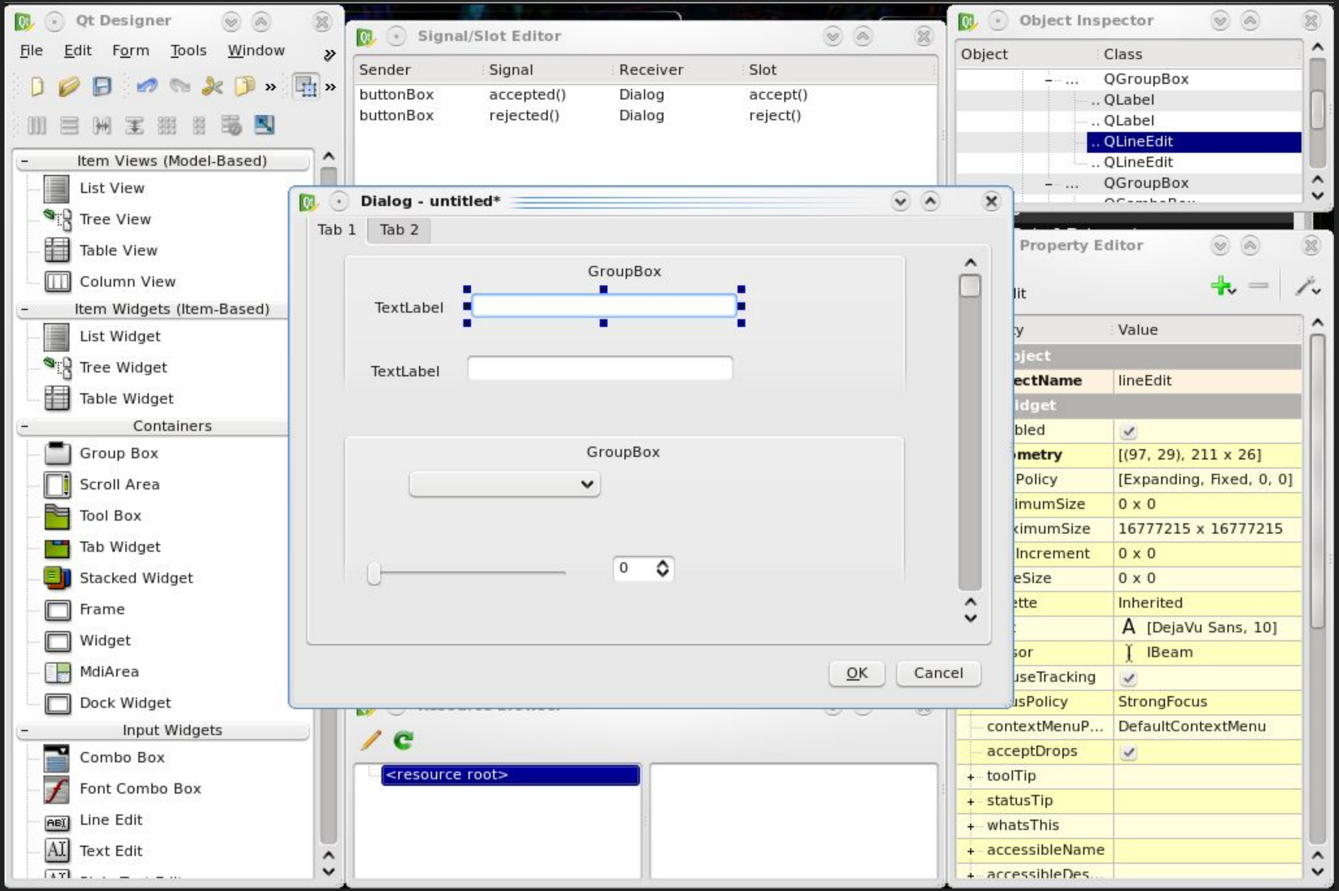
https://www.gtk.org/



GUI 框架

QT: KDE Plasma

https://qt-project.org/



名詞解釋

- dotfiles: 使用者 home 資料夾下 . 開頭的各個檔案 .config .local/share
- shell: 這裡是指覆蓋在視窗管理器／compositor 之上的桌面介面層
  - Quickshell: https://quickshell.org/
  - Noctalia Shell: https://noctalia.dev/ (本次使用)
  - DankMaterialShell: https://danklinux.com/

實作 Ricing !!!

- https://github.com/catppuccin
- https://www.reddit.com/r/unixporn
- https://wallhaven.cc
- https://docs.noctalia.dev/v4/

https://devvillie.me/blog/hackersir-nix/